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UNIVERSITY INSTITUTE OF ENGINEERING
A MINOR PROJECT ON
ON
DESIGN, FABRICATION AND PERFORMANCE ANALYSIS
OF AN AIR CUSHION VEHICLE

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS
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DECLARATION

We hereby declare that the Minor Project Report entitled

“DESIGN, FABRICATION AND PERFORMANCE ANALYSIS OF AN AIR CUSHION VEHICLE

submitted by us is an original work carried out under the guidance of Mr. Vinay Shah during the academic session 2025–26.

We further declare that this project has not been submitted previously, either in full or in part, for the award of any degree, diploma, or certificate.

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ABSTRACT

The project titled “**Design and Development of RC Hovercraft. Using 3D Printing**” focuses on the design, fabrication, and testing of a compact remote-controlled hovercraft prototype.

A hovercraft is a vehicle that moves on a cushion of air, which reduces friction between the body and the ground surface. This allows smooth movement over different surfaces such as tiles, wood, concrete, and shallow water.

We used two BLDC motors in the model. One motor creates lift by producing an air cushion below the hovercraft, while the second motor provides forward thrust for movement. Steering is controlled using a rudder connected to a servo motor and RC control system.

We designed the hovercraft body in CAD software and printed the parts using a 3D printer with PLA material. The use of 3D printing provided accurate dimensions, lightweight structure, and easy assembly of parts.

We tested the prototype on smooth surfaces, and it was able to generate lift, move forward, and respond to steering commands successfully.

This project helped us understand practical applications of aerodynamics, mechanical design, electronics integration, manufacturing processes, and control systems. The developed hovercraft can be further improved by adding autonomous navigation, surveillance features, rescue applications, and other educational uses.

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CHAPTER 1: INTRODUCTION

1.1 INTRODUCTION TO HOVERCRAFT

A hovercraft is a unique type of vehicle that moves on a cushion of air created below its body. This layer of air reduces direct contact with the ground or water surface, which helps in lowering friction and allows smooth movement.

Because of this special feature, a hovercraft can travel on different surfaces such as land, water, mud, snow, grass, and smooth floors. This multi-surface capability makes it useful for rescue operations, military transport, coastal patrolling, tourism, and research purposes.

The working principle of a hovercraft is based on lift and air pressure. A high-speed fan or propeller pushes air below the vehicle body, where it gets trapped inside a flexible skirt. The trapped air lifts the craft slightly above the surface and helps it move easily.

In our project, we designed and developed a compact RC (Remote Controlled) hovercraft using 3D printing technology. This method helped us manufacture accurate, lightweight, and low-cost parts in less time.

We used two BLDC motors in the model. One motor was used to generate lift, while the second motor was used to provide forward thrust for movement. Steering was controlled through a rudder mechanism connected to a servo motor.

The main aim of this project was to build a small working prototype that could demonstrate lift, thrust, balance, and direction control of a hovercraft system.

Through this project, we also gained practical knowledge of mechanical design, electronics integration, aerodynamics, fabrication methods, and teamwork in a simple and effective way.

1.2 OVERVIEW OF HOVERCRAFT

Hovercraft technology is based on the idea of reducing surface friction by using a cushion of air. Instead of moving directly on wheels or floating deeply in water, a hovercraft travels slightly above the surface. This gives it the ability to move smoothly over land and water.

The first successful hovercraft was developed by Sir Christopher Cockerell during the 1950s. After that, this technology became useful in many practical fields such as passenger transport, rescue operations, military services, coastal patrolling, and tourism.

Hovercrafts are generally classified into full-scale transport hovercrafts and small RC model hovercrafts. Large hovercrafts are used for carrying people and goods, while small models are mainly used for learning, research, and prototype development.

Compared with ordinary wheeled vehicles, hovercrafts can travel on mud, sand, snow, grass, and shallow water where normal vehicles may face difficulty. This makes them suitable for multi-surface transportation.

A hovercraft mainly consists of lift system, thrust system, body frame, skirt, steering rudder, and control electronics. Proper balance between these systems is important for stable movement.

In our project, we selected the hovercraft concept because it combines mechanical design, electronics, and aerodynamics in one practical model. It also provides a good platform for testing modern fabrication methods such as 3D printing.



Figure 1.1 Modern Hovercraft Vehicle

1.3 Need of Project

The need for this project arises from the importance of understanding practical engineering concepts through a working model. A hovercraft combines mechanical structure, electronics, aerodynamics, and control systems in a single application, which makes it a useful learning project for students.

In classroom study, we mostly learn theory, but a practical project helps in understanding how different components work together in real conditions. By developing an RC hovercraft, concepts such as lift generation, thrust, stability, balance, and steering can be studied more clearly.

Another need of this project is to explore modern manufacturing methods such as 3D printing. It allows parts to be designed accurately, produced quickly, and modified easily. This reduces fabrication time and overall cost.

Small hovercraft models can also be useful for testing surface movement on land or water. Such systems may be further developed in future for surveillance, rescue support, inspection work, and educational demonstrations.

we selected this project because it gives practical knowledge, encourages innovation, and helps in improving design, assembly, and problem-solving skills in an effective way.



Figure 1.2 Practical Application of Hovercraft

1.4 Problem Statement

Normal vehicles can move easily on roads, but they often face difficulty on mud, sand, water, or uneven surfaces. In such areas, wheels lose grip and movement becomes slow or unstable.

A hovercraft is a better option for these conditions because it moves on a cushion of air. Since it stays slightly above the surface, friction becomes less and movement becomes easier.

While planning this project, we understood that making a small hovercraft model is not very simple. The body needed to be lightweight, balanced, and strong enough to hold all the components properly.

We also faced the challenge of selecting suitable motors, battery, and propellers that could provide enough lift and forward thrust. If the power is low, the hovercraft may not perform properly.

Another important issue was steering control. The rudder had to be placed correctly so that the model could turn smoothly during movement.

The main problem taken in this project was to design and build a compact RC hovercraft that is low-cost, stable, and able to generate lift, move forward, and turn effectively.

1.5 Objectives of the Project

The main objective of this project was to design and develop a small RC hovercraft that could demonstrate the practical working principle of an air cushion vehicle. We selected this project to gain hands-on experience in design, fabrication, electronics, and control systems.

One of the important objectives was to create a lightweight body structure that could generate proper lift and move smoothly on flat surfaces. For this purpose, we used 3D printing technology to make accurate and low-cost parts.

Another objective was to use suitable electronic components such as BLDC motors, ESC, battery, servo motor, and RC controller for proper lift, thrust, and steering operation.

We also wanted to understand how weight distribution, component placement, and air flow affect the stability and performance of the hovercraft during movement.

This project was also aimed at improving our practical knowledge in CAD modelling, assembly work, wiring, testing, and troubleshooting.

Overall, the objective of this project was not only to build a working model, but also to learn how mechanical and electronic systems work together in a real application.

1.6 Scope of Project

The scope of this project is to build and test a small RC hovercraft that can move smoothly using the principle of an air cushion. Through this model, we aimed to combine design, electronics, fabrication, and control systems in one practical project.

This project is useful for students because it helps in understanding how lift, thrust, steering, and balance work in real conditions. Instead of learning only theory, the working model gives practical knowledge and hands-on experience.

The hovercraft can also be used for academic displays, technical exhibitions, and project demonstrations. It is a good example of how mechanical and electronic systems can work together in a simple model.

There is also future scope for improving this project by adding features such as sensors, camera module, GPS system, Bluetooth control, or Wi-Fi operation. These upgrades can make the model more advanced and efficient.

In future, similar hovercraft systems may be useful in rescue work, surface inspection, surveillance, and movement in flooded or difficult areas.

Overall, this project gives wide scope for learning, innovation, and further development in the field of mechatronics engineering.



Figure 1.3 Future Scope of Hovercraft in Rescue Operations

1.7 Advantages of Hovercraft

Hovercraft offers several advantages over conventional vehicles because it moves on a cushion of air instead of directly touching the surface. Due to this feature, friction is greatly reduced, which results in smoother movement and better efficiency.

One of the main advantages of a hovercraft is its ability to travel on different surfaces such as land, water, mud, sand, snow, grass, and smooth floors. Unlike normal vehicles, it does not require a fixed road or track.

Hovercraft can also move easily in shallow water and flooded areas where ordinary vehicles may face problems. Because of this, they are highly useful in rescue, emergency, and disaster relief operations.

Another important advantage is high maneuverability. The craft can change direction quickly with the help of rudder control and can move freely in open spaces.

The structure of a hovercraft can be lightweight, especially when modern materials and 3D printing methods are used. A lightweight body improves speed and reduces power consumption.

Hovercraft models are also very useful for educational purposes. They help students understand practical concepts such as aerodynamics, lift generation, thrust control, motor operation, battery systems, and mechatronics.

For small RC hovercraft models, maintenance is easier because fewer moving contact parts are used in the system.

• Main advantages of hovercraft are listed below:

1. Low friction movement
2. Multi-surface travel capability
3. Operation on land and water
4. Lightweight construction
5. High maneuverability
6. Useful in rescue operations
7. Modern transportation technology

1.8 Limitations of Hovercraft

Even though a hovercraft has many useful features, it also has some limitations in practical use. One of the main limitations is that it needs continuous power to maintain the air cushion below the body. If the power supply becomes low, the lifting effect also reduces.

A hovercraft performs better on smooth and flat surfaces. On rough ground, rocky areas, or places with obstacles, its movement may become unstable and less efficient.

Another limitation is sensitivity to strong winds. Since the craft remains slightly above the surface, wind can affect its direction, balance, and control during operation.

The system also creates noticeable noise because of the high-speed fan, propeller, or motors used for lift and thrust generation. This may create discomfort in some surroundings.

Steering control of a hovercraft is comparatively more difficult than ordinary wheeled vehicles. Proper balancing and control are necessary for smooth movement.

For RC hovercraft models, battery backup is limited. The vehicle can operate only for a certain time before the battery needs recharging.

Regular maintenance is also important, especially for the skirt, propeller, and electronic components. Any damage or air leakage can reduce overall performance.

- Main limitations of hovercraft are listed below:

1. Requires continuous power supply
2. Better performance only on smooth surfaces
3. Strong wind affects control and balance
4. Produces noise during operation
5. Steering is comparatively difficult
6. Limited battery backup in RC models
7. Regular maintenance is necessary

1.9 Chapter Summary

In this chapter, we discussed the basic concept of a hovercraft and its working principle. We explained how a hovercraft moves on a cushion of air and can travel on different surfaces with less friction.

The need for selecting this project was also discussed along with the practical problems that can be solved using this technology. The main objectives and scope of the project were presented to show the learning value and future possibilities of the model.

We also studied the major advantages of hovercraft such as multi-surface movement and low friction operation, along with some limitations like power requirement, wind effect, and control challenges.

Overall, this chapter provided a clear introduction to the project and created a strong base for the next chapters of design, components, fabrication, and testing.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

Before starting the development of our RC hovercraft, it was important to study previous hovercraft designs, available technologies, and working methods. A literature review helps in understanding how similar systems were developed and what improvements can be made in a new project.

By studying earlier hovercraft models, we learned about lift generation, thrust systems, steering mechanisms, material selection, and body balancing. It also helped us understand common problems such as air leakage, instability, and power consumption.

Different types of hovercraft systems are used today for transportation, rescue work, military operations, tourism, and research purposes. Small RC hovercraft models are also used for learning and experimental applications.

For our project, we studied various compact hovercraft designs and selected a simple, lightweight, and low-cost model that could be developed using 3D printing technology.

This chapter includes the history of hovercraft, existing systems, RC hovercraft models, comparison of technologies, and research gap related to the present project.

2.2 History of Hovercraft

The idea of hovercraft was developed to create a vehicle that could move over land and water with minimum friction. Instead of depending on wheels or deep water flotation, the vehicle uses a cushion of air for movement.

The modern hovercraft concept is mainly linked with Sir Christopher Cockerell, a British engineer, who worked on practical air cushion vehicle technology during the 1950s. His experiments showed that controlled air pressure below a vehicle could generate lift and improve movement.

The first successful practical hovercraft, known as SR.N1, was introduced in 1959. It became famous after crossing the English Channel, proving that hovercraft technology could be used for transportation.

After this achievement, hovercrafts were developed for many applications such as passenger transport, rescue services, coastal patrolling, cargo movement, and military use.

With time, hovercraft designs improved because of better engines, lightweight materials, stronger skirts, and modern control systems. These developments made hovercrafts more efficient and reliable.

Today, hovercraft technology is also used in small RC models for educational projects, hobby applications, and research work. These compact models help students understand the basic working principle of lift, thrust, and steering.

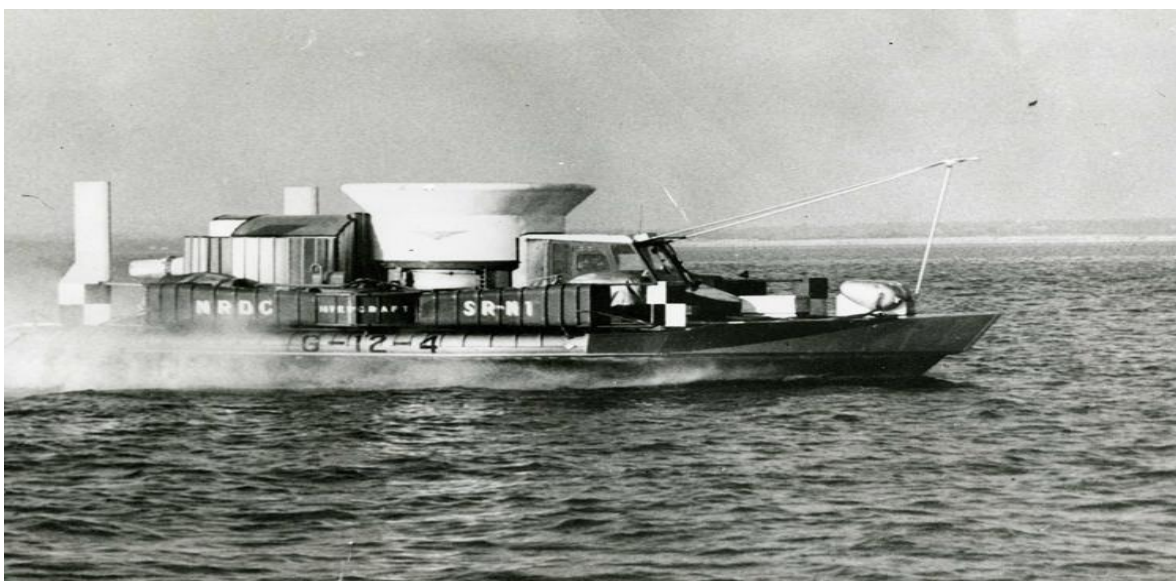


Figure 2.2 First Practical Hovercraft SR. N1

2.3 Existing Hovercraft Systems

At present, hovercraft systems are used in different sizes and designs according to their purpose. Some hovercrafts are large and used for carrying passengers or goods, while some are small models used for learning and testing purposes.

Large hovercrafts are commonly used in coastal areas, river crossings, and shallow water regions where normal boats or road vehicles may face difficulty. They are useful because they can move over both land and water surfaces.

Hovercrafts are also used in rescue operations during floods or emergency situations. Since they can travel over water, mud, and uneven surfaces, they are considered suitable for difficult conditions.

In defence applications, hovercraft systems are used for coastal patrolling, transport of equipment, and movement in marshy areas. Their fast movement and multi-surface ability make them useful in such operations.

Small hovercraft models are widely used in schools, colleges, and research labs. These models are generally powered by electric motors and batteries, and they help in understanding lift generation, thrust, steering, and balance.

While studying these existing systems, we found that many large hovercrafts are costly and complex, whereas small RC models are more suitable for student projects and practical learning.



Figure 2.3 Existing Hovercraft Transportation System

2.4 RC Hovercraft Models

RC hovercraft models are small versions of hovercraft that can be controlled from a distance using a transmitter and receiver system. These models are commonly used in student projects, hobby work, experiments, and learning activities.

The basic working principle of an RC hovercraft is the same as a full-size hovercraft. Air is forced below the body to create a cushion that lifts the craft slightly above the surface. Another motor is used to generate forward thrust for movement.

In most RC hovercraft models, electric motors are preferred because they are lightweight, efficient, and easy to manage. Rechargeable batteries such as LiPo batteries are generally used to supply power to the system.

Direction control is usually achieved by using a rudder connected to a servo motor. With the help of a handheld remote controller, the user can control movement and turning of the craft.

These models are very useful for understanding practical engineering concepts such as lift force, thrust generation, wireless control, balancing, and lightweight design.

During our study of different RC hovercraft models, we found that compact electric systems are more suitable for student projects because they are affordable, simple to build, and easy to maintain.

Our project is based on the same concept, where we designed a compact RC hovercraft using BLDC motors, electronic control components, and 3D printed parts.



Figure 2.4 Example of RC Hovercraft

2.5 Comparison of Existing Technologies

Different technologies are available for making small transport models such as wheeled vehicles, tracked systems, boats, drones, and hovercrafts. Each system works in a different way and is useful for specific conditions.

Wheeled vehicles are simple in design, economical, and easy to control. They perform well on roads and smooth surfaces, but they may face difficulty on mud, water, sand, or uneven ground.

Tracked vehicles provide better grip and stability on rough surfaces. They can move on soft soil and difficult terrain, but they are heavier and usually consume more power because of higher friction.

Boats are mainly used for movement on water surfaces. They work efficiently in rivers, lakes, and ponds, but they cannot travel on land.

Drones are useful for aerial movement and surveillance. They can reach places quickly, but they are more expensive, need skilled control, and usually have limited battery time.

Hovercrafts are different from all these systems because they can move on both land and water using a cushion of air. They require less surface contact, which helps in smoother movement with lower friction.

After studying these technologies, we selected the hovercraft concept for our project because it provides multi-surface movement, interesting design, and good learning opportunities in mechanics and electronics.

Technology	Surface Capability	Complexity	Educational Value
Wheeled Vehicle	Road Only	Low	Medium
Tracked Vehicle	Rough Terrain	Medium	Medium
Boat	Water Only	Medium	Low
Drone	Air	High	High
Hovercraft	Land + Water	Medium	High

2.6 Research Gap

During our study of different hovercraft systems, we found that many available models are either large commercial vehicles or very simple hobby models. Large hovercrafts are costly, complex in design, and not practical for student projects or small-scale fabrication.

Many small RC hovercraft models available online are mainly made for entertainment purposes. They often use simple body structures and may not focus on proper balance, durability, or engineering improvements.

We also noticed that several student projects use materials such as foam, cardboard, or plastic sheets. These materials are lightweight, but they may not provide accurate shape, good strength, or long-term durability.

Another gap observed was the limited use of modern manufacturing methods in low-cost student models. Many designs do not make proper use of 3D printing, which can help in producing accurate, lightweight, and customized parts.

Some models are able to generate lift, but steering and forward movement are not properly optimized. In other cases, the design works, but balancing and stability remain weak.

Because of these gaps, there is a need for a compact, low-cost, and better-designed RC hovercraft that uses 3D printed parts, suitable electronic components, and proper control systems.

Our project was planned to address this need by developing a student-friendly hovercraft model that is practical, easy to assemble, and useful for learning purposes.

2.7 Summary

In this chapter, we reviewed the background and development of hovercraft technology. We studied how hovercraft systems were introduced, how they improved over time, and where they are used today.

We also looked at different existing hovercraft systems, including large transport hovercrafts and small RC models used for learning and experimental purposes. This helped us understand the common design features and working methods of such systems.

A comparison with other technologies such as wheeled vehicles, tracked systems, boats, and drones was also made. From this comparison, we found that hovercrafts offer the special advantage of moving on both land and water with less surface friction.

During the literature review, we also identified certain gaps in existing student models, such as weak body design, limited control systems, and less use of modern fabrication methods.

Based on these observations, we selected a compact and practical hovercraft design for our project using 3D printing and suitable electronic components.

Overall, this chapter gave us useful knowledge and helped us plan the next stage of design, fabrication, and testing of the RC hovercraft model.

CHAPTER 3: DESIGN AND METHODOLOGY

3.1 Introduction

After studying the background and existing hovercraft systems, the next step was to design a practical model for our project. The aim was to create a compact RC hovercraft that would be lightweight, stable, and easy to control.

During the design stage, we focused on important factors such as body dimensions, weight distribution, component placement, motor mounting, and steering arrangement. Proper planning was necessary because even small changes in balance can affect hovercraft performance.

We selected a compact structure so that the model could be easy to carry, simple to assemble, and suitable for indoor and outdoor testing on smooth surfaces.

For better accuracy and clean fabrication, we used CAD software to design the main body and other required parts. This also helped us modify the design whenever needed before printing.

After finalizing the model, the parts were manufactured using 3D printing technology with PLA material. This method made it possible to produce lightweight and low-cost parts with proper dimensions.

This chapter explains the complete design process, dimensions, material selection, printing method, and assembly steps used for developing the hovercraft model.

Figure 3.1 CAD Design of RC Hovercraft

3.2 Final Dimensions

After completing the design and fabrication of the hovercraft, the final dimensions were obtained from the CAD model and printed parts. The prototype was developed in a compact form so that it remains lightweight, easy to carry, and convenient for practical testing and demonstrations.

The approximate overall dimensions of the hovercraft are **150.1 mm** in length, **238.6 mm** in width, and **151.3 mm** in height. These dimensions were selected carefully to achieve a proper balance between **size, weight, stability, and performance**. Proper dimensions were important because even small changes in size can affect lift generation and movement.

The body size provides sufficient space for installing important components such as **BLDC motors, battery, ESC, servo motor, receiver, and wiring connections**. The internal arrangement of these components was planned in such a way that the weight remains balanced during operation.

At the same time, the compact structure helps in reducing material consumption and keeping the total weight low. A lightweight body improves lift performance and reduces extra load on the motors.

Because of its lightweight design, the hovercraft can generate lift more effectively and move smoothly over flat surfaces. Its moderate size also makes handling, transportation, and storage easier during testing.

Overall, the final dimensions of the developed hovercraft were found suitable for stable movement, better control, and successful demonstration of hovercraft operating principles.

3.3 Concept Selection

Before starting the detailed design, different concepts were studied for developing the hovercraft model. Various body shapes, propulsion methods, and steering arrangements were considered to select a suitable design for the project.

During the initial stage, simple rectangular and circular body concepts were reviewed. However, these designs were not found suitable because they either increased weight, reduced stability, or provided less space for component mounting.

After comparing different options, a streamlined rectangular body with rounded front edges was selected. This shape was found better for maintaining balance, reducing air resistance, and providing enough internal space for electronic components.

For lift generation, an electric motor with propeller was selected to create an air cushion below the body. A separate rear motor was chosen for forward thrust so that lift and movement could be controlled effectively.

For steering control, a rudder mechanism connected to a servo motor was selected. This arrangement allows easy left and right turning through remote control operation.

The final concept was selected because it was compact, lightweight, simple to fabricate, and suitable for 3D printing technology.

Overall, the selected concept provided a practical and efficient base for developing the RC hovercraft model successfully.

3.4 CAD Modelling

After selecting the final concept, the next step was to prepare the 3D design of the hovercraft using CAD software. Computer Aided Design helped us create accurate models of all required parts before fabrication.

The complete structure of the hovercraft was divided into separate components such as the main body, skirt frame, rear motor housing, and rudder. Designing the parts separately made modification, assembly, and printing easier.

During modelling, proper attention was given to dimensions, wall thickness, mounting holes, and component placement. Sufficient space was provided for motors, battery, ESC, servo motor, receiver, and wiring connections.

The CAD model also helped in checking alignment of the rear motor housing and rudder system before printing. This reduced chances of fitting errors during final assembly.

Several small changes were made during the design stage to improve balance, strength, and overall appearance of the hovercraft. CAD software made these changes easy and quick.

After completing the design, all parts were exported in STL format for 3D printing. This step connected the digital design process with physical manufacturing.

Overall, CAD modelling played an important role in developing an accurate, lightweight, and practical hovercraft prototype.

3.5 Main Body Design

The main body is the most important structural part of the hovercraft because all major components are mounted on it. It acts as the base of the complete system and helps in maintaining balance, strength, and stability during operation.

While designing the main body, our first priority was to keep it lightweight so that the hovercraft could generate lift easily. At the same time, the body needed to be strong enough to support motors, battery, and other electronic parts safely.

The final dimensions of the main body were approximately **150.1 mm** in length, **238.6 mm** in width, and **58.3 mm** in height. These dimensions were selected to provide enough internal space for component mounting while keeping the body compact.

The front side of the body was designed with a curved profile to improve appearance and allow smoother airflow during movement. The rear side was prepared for fitting the motor housing and rudder mechanism.

The complete body was designed in CAD software and manufactured using PLA material through 3D printing technology.

In the end, the final body design was lightweight, strong, and suitable for smooth operation of the hovercraft model.

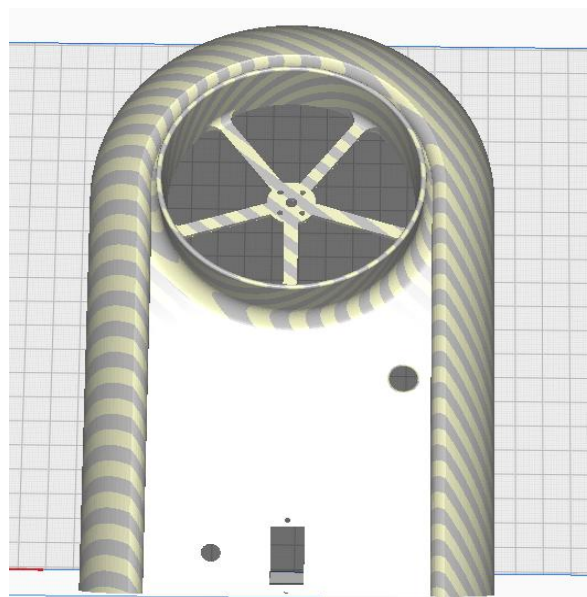


Figure 3.1 Main Body Design of Hovercraft

3.6 Skirt Frame Design

The skirt frame is an important lower structural part of the hovercraft. It supports the bottom section of the body and helps in maintaining the air cushion area required for lift generation. It also provides proper support for fixing the skirt material securely.

While designing the skirt frame, the main objective was to keep it lightweight and strong. Since this part is placed at the lower side of the hovercraft, unnecessary weight was avoided to improve lift performance.

The final dimensions of the skirt frame were approximately **150.1 mm** in length, **238.6 mm** in width, and **3.0 mm** in height. These dimensions were selected according to the size of the main body so that both parts could fit properly during assembly.

The frame was designed with an open inner section to reduce material usage and overall weight. At the same time, sufficient support ribs were provided to maintain strength and rigidity.

Proper alignment with the main body was also considered during design so that air leakage could be minimized and stable lift could be achieved.

The skirt frame was designed in CAD software and fabricated using **PLA material** through 3D printing technology.

Overall, the final skirt frame design was lightweight, strong, and suitable for supporting efficient hovercraft operation.

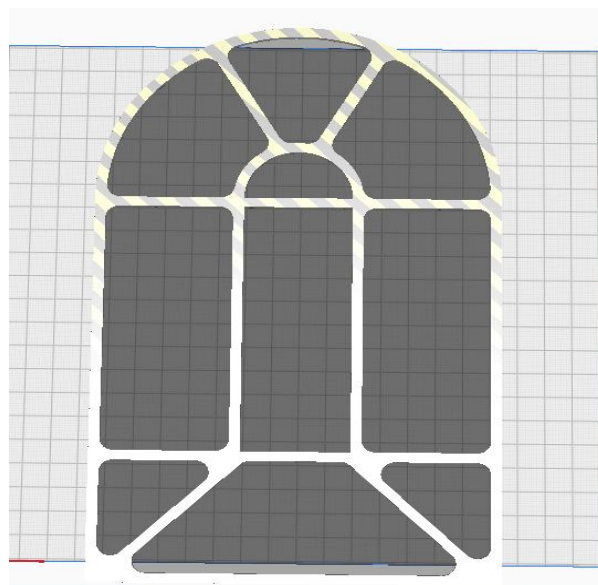


Figure 3.2 3D Printed Skirt Frame Design

3.7 Rear Motor Housing Design

The rear motor housing is an important part of the hovercraft because it holds the thrust motor at the back side of the model. It provides proper support to the motor and helps in keeping the motor fixed in the correct position during operation.

While designing this part, our main focus was to make it strong, lightweight, and properly aligned with the body. Correct alignment was necessary so that the thrust produced by the motor moves the hovercraft in the forward direction without unnecessary vibration.

The final dimensions of the rear motor housing were approximately **95.0 mm** in length, 35.0 mm in width, and **93.0 mm** in height. These dimensions were selected according to the motor size and the rear body structure.

The housing was designed with a curved protective shape to support the motor safely and improve the overall appearance of the hovercraft. It also helped in guiding airflow from the propeller in the required direction.

Proper mounting sections were included so that the housing could be attached easily to the main body after printing. This reduced fitting problems during final assembly.

The complete part was designed in CAD software and produced using **PLA material** through 3D printing technology.

Overall, the rear motor housing design was found stable, lightweight, and suitable for smooth thrust generation in the hovercraft model.

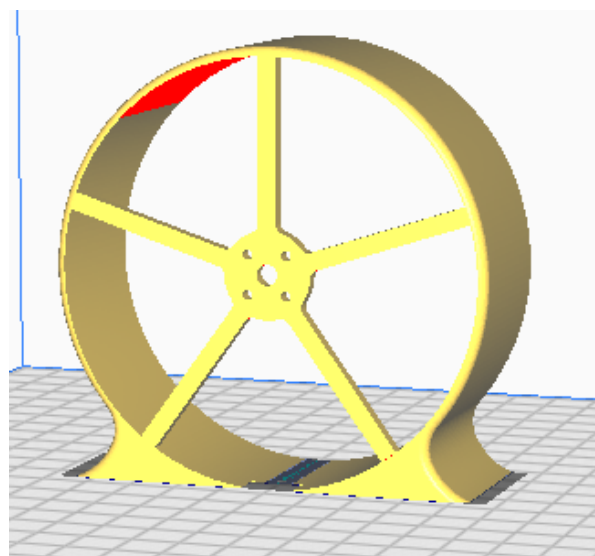


Figure 3.3 CAD Model of Rear Motor

3.8 Rudder Design

The rudder is one of the main control parts of the hovercraft and is used for steering during movement. It was placed behind the rear thrust motor so that the airflow from the propeller could be used to turn the hovercraft left or right whenever required.

While designing the rudder, our main focus was to keep it light in weight but strong enough for regular operation. Since the rudder is connected to a servo motor, a lighter design helps in faster movement and reduces extra load on the steering mechanism.

The final dimensions of the rudder were approximately **75.0 mm** in length, **73.1 mm** in width, and **5.2 mm** in height. These dimensions were selected to provide enough surface area for controlling airflow while keeping the part compact.

The shape of the rudder was kept simple so that air could pass smoothly and steering response could remain effective. Proper holes and mounting points were also provided for easy connection with the servo linkage.

During assembly, correct placement of the rudder was very important because even a small alignment error could affect turning performance and control.

The complete rudder was designed in CAD software and later manufactured using **PLA material** through 3D printing.

Overall, the final rudder design was found lightweight, easy to operate, and suitable for smooth directional control of the hovercraft model.

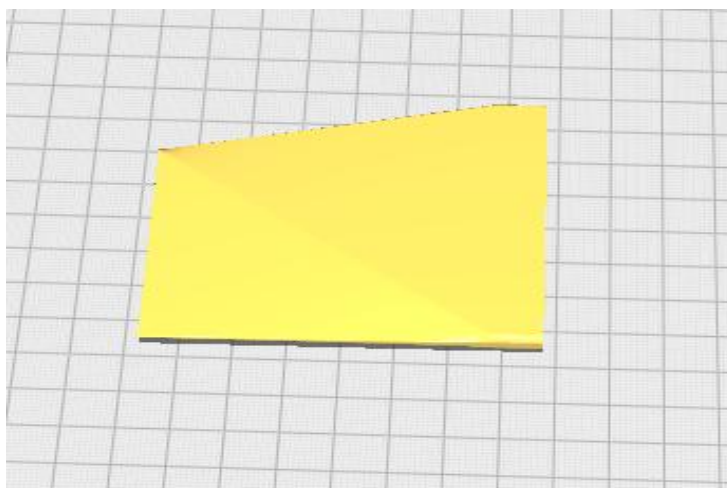


Figure 3.4 CAD Model of Single Rudder

3.9 Material Selection

Material selection was an important part of this project because the performance of the hovercraft depends greatly on its total weight, strength, and durability. If the body is too heavy, lift generation becomes difficult, and if the material is weak, the structure may not remain stable during operation.

While planning the model, we needed a material that was lightweight, easy to shape, affordable, and suitable for 3D printing. After comparing common options, **PLA** was selected for manufacturing the major parts of the hovercraft.

One of the main reasons for choosing **PLA** was its low weight. A lighter structure helps the hovercraft rise more easily on the air cushion and reduces extra load on the motors.

PLA also provides enough strength for small components such as the main body, skirt frame, motor housing, and rudder. It can hold mounted parts properly and is suitable for prototype models.

Another reason for selecting **PLA** was that it is easy to print and gives good dimensional accuracy. This helped us in getting better fitting between different parts during final assembly.

PLA is also easily available in the market and its cost is reasonable, which makes it a practical choice for student projects and experimental work.

Overall, PLA was found to be a balanced option because it offers low weight, sufficient strength, easy printing, and good cost effectiveness for the hovercraft model.



Figure 3.5 PLA Filament Used for 3D Printed Parts

3.10 3D Printing Process

After completing the CAD design of all required parts, the next step was to manufacture them using 3D printing technology. This method was selected because it can produce custom parts accurately with less material waste and lower cost.

The final CAD models were first converted into **STL files** and then imported into slicing software. In the slicer, settings such as layer height, infill percentage, print speed, and support generation were adjusted according to the shape and size of each part.

Before starting the printing process, the printer bed was cleaned and leveled properly to achieve good first-layer adhesion. Proper bed leveling was necessary because it directly affects print quality and dimensional accuracy.

Each hovercraft part, including the main body, skirt frame, rear motor housing, and rudder, was printed separately. Printing the components one by one made handling and final assembly easier.

During printing, the process was checked regularly for possible issues such as poor adhesion, layer shifting, stringing, or warping. Small corrections were made whenever needed to maintain print quality.

After printing was completed, all parts were removed carefully from the build plate. Support material and rough edges were cleaned so that the parts could fit properly during assembly.

By using 3D printing, we were able to develop lightweight, accurate, and low-cost parts in less time. This process helped us convert the digital design into a practical hovercraft prototype successfully.

3.11 Print Settings

Selecting proper print settings was an important step during fabrication because the quality of printed parts depends greatly on these parameters. Good settings helped us obtain parts with proper strength, smooth finish, and accurate dimensions.

While printing the hovercraft parts, a medium layer height was selected so that print quality remained good and the printing time also stayed reasonable. Very small layer height increases print time, while very high layer height can reduce surface finish.

The infill percentage was chosen according to the purpose of each part. Parts that required more strength, such as the main body and motor housing, were printed with moderate infill. Lighter parts were printed with lower infill to reduce unnecessary weight.

Print speed was kept at a balanced value to avoid vibration, poor finishing, or dimensional errors. A stable speed helped in producing cleaner and more reliable parts.

Support material was used only where needed, especially for curved surfaces and overhanging areas. This reduced extra material usage and made post-processing easier.

The nozzle and bed temperatures were adjusted according to PLA material requirements to ensure smooth extrusion and good first-layer adhesion.

During printing, we also made a few small adjustments after observing the output quality. This helped in improving the final result of the printed components.

With proper print settings, we were able to produce strong, lightweight, and accurately fitted parts for the hovercraft assembly.

3.12 Assembly Process

After completing the printing of all required parts, the next stage of the project was the assembly of the complete hovercraft model. All printed components were first checked carefully for proper dimensions and fitting. Support material, rough edges, and extra plastic were removed so that the parts could be joined smoothly.

The assembly process started with the main body, which acted as the base structure of the hovercraft. The skirt frame was mounted on the lower side of the body and aligned properly. This step was important because correct fitting of the lower frame helps in maintaining the air cushion area.

After fixing the lower frame, the rear motor housing was attached to the back side of the main body. It was mounted firmly so that the thrust motor could remain stable during high-speed rotation.

The lift motor and propeller arrangement were then installed in their required positions. Proper mounting was ensured to avoid vibration and power loss during operation. The rear thrust motor was also fitted carefully inside the motor housing.

The rudder was mounted behind the rear motor and connected to the servo motor through linkage. Rudder movement was checked manually to confirm smooth left and right turning.

After mechanical assembly, electronic components such as the **ESC, battery, receiver, and wiring connections** were placed inside the body. Components were arranged carefully to keep the weight balanced on all sides.

All wire connections were checked properly and moving parts were tested before powering the system. Loose screws and misalignment, if found, were corrected at this stage.

After complete assembly, the hovercraft was ready for final testing and performance evaluation.

3.13 BLOCK DIAGRAM

The block diagram of the RC hovercraft system is shown below. It explains the basic working arrangement of the complete model in a simple way. It shows how signals are sent from the transmitter to the receiver and then passed to the ESC units, motors, and servo motor. It also shows how power is supplied from the LiPo battery to the required components. With the help of this diagram, the connection and working of all major parts can be understood easily.

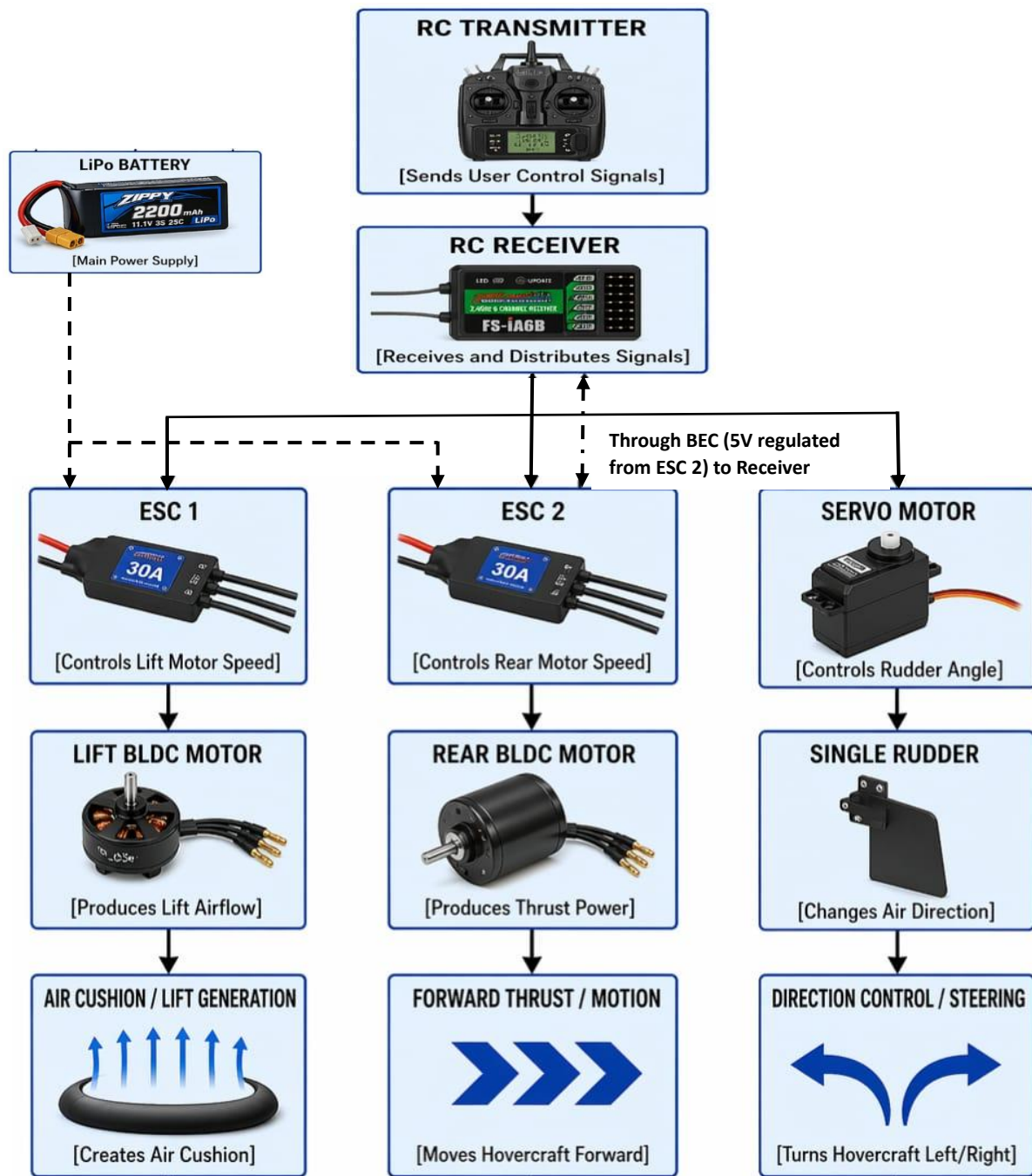


Figure 3.1 RC Hovercraft System Block Diagram

LEGEND

-  Solid Arrow = Control Signal
-  Dashed Arrow = Power Supply

3.14 Software Used

Different software tools were used during the hovercraft project for design, fabrication, and documentation work.

FreeCAD was used for CAD modelling of parts such as the main body, rudder, motor housing, and skirt frame. It helped in preparing accurate 3D models before manufacturing.

Ultimaker Cura version 5.10.2 was used for slicing the designed models and generating G-code for 3D printing. Print settings such as layer height, infill, and support were adjusted in this software.

Microsoft Word was used for preparing the project report, arranging chapters, inserting images, formatting tables, and final documentation.

These software tools were helpful in completing the project work in a proper and systematic way.

3.15 Chapter Summary

In this chapter, we discussed the complete design and development process of the hovercraft model. Different ideas were studied in the beginning, and the most suitable concept was selected on the basis of lightweight structure, stability, and practical working.

The final dimensions of the hovercraft were decided after considering component space, balance, and smooth operation. CAD software was used to create accurate designs of the main body, skirt frame, rear motor housing, and rudder before fabrication.

We also planned the placement of motors, battery, ESC, receiver, and steering components so that the model remains balanced during movement. Proper attention was given to air flow and overall body structure for better performance.

For fabrication, 3D printing technology was selected because it allowed us to make lightweight and accurate parts at low cost. It also made assembly and future modifications much easier.

Finally, this chapter explained how the hovercraft concept was converted into a practical working prototype through proper design, planning, and manufacturing steps.

CHAPTER 4: COMPONENTS AND HARDWARE USED

4.1 Introduction

After completing the design and fabrication of the hovercraft body, the next step was to choose the components required for running and controlling the model. Proper selection of components was very important because the performance of the hovercraft depends on power, weight, control response, and reliability.

The hovercraft needed both mechanical and electronic parts to perform three main functions: lift generation, forward movement, and steering control. Because of this, each component was selected carefully according to the size and requirements of the prototype.

The major components used in this project were BLDC motors, Electronic Speed Controllers (ESC), propellers, LiPo battery, servo motor, RC transmitter and receiver, wires, connectors, and mounting accessories.

While choosing these parts, we considered important factors such as low weight, sufficient power, compatibility, battery efficiency, and easy installation. Lightweight components were preferred because they help the hovercraft generate lift more easily.

Reliable electronics were also necessary for smooth and safe operation during testing. If unsuitable components are used, the hovercraft may face poor control, low performance, or unstable movement.

During planning, we compared different available options and selected parts that were practical, affordable, and suitable for a student project.

This chapter describes the important components used in the hovercraft, their functions, and the reasons for selecting them for the final model.

4.2 BLDC Motor

BLDC motor stands for Brushless DC motor. It is one of the most important components used in our hovercraft project. These motors are commonly used in RC models because they provide high speed, better efficiency, and reliable performance with low maintenance.

In our project, two BLDC motors were used. The first motor was a 2212 1000KV motor, which was selected for lift generation. It was used to push air below the hovercraft body and create an air cushion for lifting the model slightly above the surface.

The second motor was a 2212 2200KV motor. This motor was used for forward thrust because it provides higher speed. It was mounted at the rear side of the hovercraft with a propeller to move the craft in the forward direction.

We selected these motors because they are lightweight, compact in size, and suitable for battery-powered RC projects. Their power output was sufficient for our hovercraft design.

During testing, we found that proper motor mounting and alignment were very important. If the motor position is not correct, lift and movement performance may be affected.

Both motors were connected to separate ESC units and powered through the LiPo battery system.

Thus, the selected BLDC motors were responsible for lift generation and forward motion, making them key components of the hovercraft system.



Figure 4.1 BLDC Motors Used for Lift and Forward

4.3 Electronic Speed Controller (ESC)

Electronic Speed Controller, commonly called ESC, is used to control the speed of BLDC motors. It receives signals from the receiver and supplies power to the motor according to the throttle command.

In our project, two ESC units were used because two BLDC motors were installed in the hovercraft. One ESC was used for the lift motor and the second ESC was used for the rear thrust motor.

We used 30A BLDC ESCs with SimonK firmware. These ESCs were selected because they matched the power requirement of our 2212 motors and gave smooth response during operation.

The ESC helps the motor start properly and allows speed control whenever needed. This was useful for controlling lift and forward movement during testing.

While testing the hovercraft, we found that ESC calibration was important. If calibration was not done properly, the motor response became slow or uneven.

Both ESC units were connected with the battery, receiver, and motors using proper wiring connections.

Thus, the ESC was an important part of the system and helped in smooth control of both motors.



Figure 4.2 ESC Used for BLDC Motor Speed Control

4.4 Propeller

The propeller is one of the important parts used with the BLDC motor in our hovercraft. Its main work is to push air when the motor rotates at high speed. This airflow is then used for lift generation and forward movement.

In our project, two propellers were used with two motors. One propeller was fitted on the lift motor to send air below the hovercraft body and create an air cushion. The second propeller was fitted on the rear motor to produce thrust and move the hovercraft forward.

We used 5045 size propellers in the project. These were selected because they matched well with the 2212 motors and gave good airflow during operation.

While testing, we found that the correct propeller size is important. A very small propeller gives less airflow, while an oversized propeller can put extra load on the motor.

We also noticed that proper balancing and tight fitting of the propeller were necessary. If the propeller becomes loose or unbalanced, vibration may occur and performance can be reduced.

Both propellers were mounted carefully on the motor shafts using proper fittings.

Thus, the propellers were useful in creating lift and forward motion, which are essential for the working of the hovercraft.



Figure 4.3 3 Inch Propeller Used for Lift and Thrust System

4.5 LiPo Battery

The LiPo battery was used as the main power source of the hovercraft model. It supplied power to both ESC units, the receiver, and the servo motor during operation. A rechargeable lithium polymer battery was selected because it is lightweight, compact in size, and gives good power output.

The battery used in this project had a rating of 11.1 V (3S) and a capacity of 2200 mAh. This was suitable for running the motors and other control components for the required time. It was also able to provide enough current during motor starting and regular working conditions.

The battery was placed on the upper section of the main body to maintain proper balance of the hovercraft. This position also made charging and replacement easier whenever needed. Proper support was given so that the battery did not move during operation.

An XT60 connector was used to connect the battery with the system and provide a safe power connection to the ESC units.

In the end, the selected LiPo battery was suitable for smooth, lightweight, and stable working of the hovercraft model.



Figure 4.4: LiPo Battery Used as Main Power Supply

4.6 Servo Motor

Servo motor is a small control motor used for accurate angular movement. It is commonly used in RC projects where controlled rotation is required for steering or movement.

In our hovercraft project, one servo motor was used for steering control. It was connected to the rudder mechanism at the rear side of the hovercraft. When commands were given from the transmitter, the servo moved the rudder left or right and changed the direction of the hovercraft.

We used an MG995 servo motor with 180-degree rotation. This servo was selected because it provides good torque, reliable movement, and is suitable for RC applications.

The servo motor was powered through the receiver and worked according to the control signals sent by the transmitter. Its response during testing was smooth and quick.

While testing, we found that proper linkage between the servo arm and rudder was important. If the connection was loose, steering movement became inaccurate.

The servo was mounted firmly on the body to avoid unwanted vibration during operation.

Thus, the MG995 servo motor played an important role in steering control and helped the hovercraft turn properly during movement.

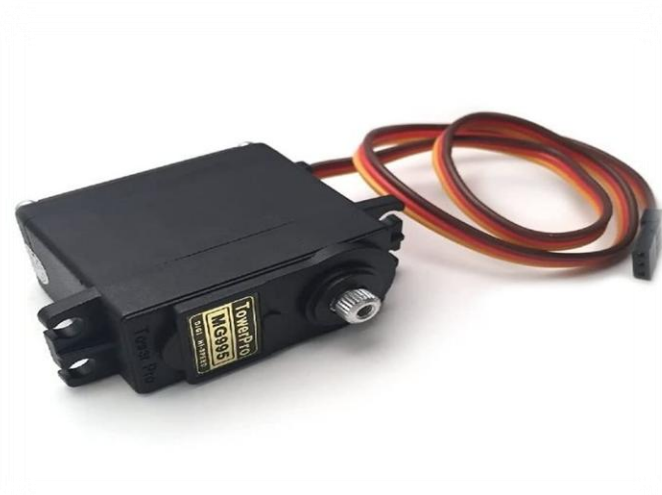


Figure 4.5 Servo Motor Used for Steering and Rudder Control

4.7 XT60 Connector

XT60 connector is a commonly used power connector in RC projects and battery-operated systems. It is known for its strong grip, safe connection, and ability to carry higher current without excessive heating.

In our hovercraft project, one XT60 male and female pair was used for connecting the LiPo battery with the main power circuit. It provided a secure connection between the battery and the electronic components.

We selected the XT60 connector because it is reliable, easy to connect and disconnect, and suitable for the power requirement of our hovercraft system.

The connector helped in smooth power transfer from the battery to the ESC units and motors. Proper connection was important because loose terminals can cause power loss or interruption during operation.

During assembly, we made sure that the wiring polarity was correct before connecting the battery. Wrong polarity can damage electrical components.

We also observed that the XT60 connector remained stable during testing and did not loosen because of vibration.

Thus, the XT60 connector was a small but important part of the electrical system and helped in safe power connection of the hovercraft.



Figure 4.6 XT60 Connector Used for Battery Power Connection

4.8 RC Control System

The RC control system was used to operate the hovercraft wirelessly from a distance. It helped us control the model without touching it directly during movement and testing.

This system mainly consists of two parts, a transmitter and a receiver. The transmitter was held by the user, while the receiver was installed inside the hovercraft body.

In our project, the transmitter was used to send commands for movement and steering. The receiver accepted those signals and passed them to the connected ESC units and servo motor.

With the help of this system, we were able to control motor speed, lift operation, forward movement, and turning direction. It made the hovercraft easier to handle during testing.

We selected the RC control system because it is simple to use, reliable, and commonly used in small RC projects.

During testing, we found that proper channel setting and good signal range were necessary for smooth operation. After pairing the system correctly, the hovercraft responded well to control inputs.

The RC system also improved safety because the model could be controlled while standing at a suitable distance.

Thus, the RC control system was an important part of the project and helped in smooth wireless operation of the hovercraft.



Figure 4.7 RC Control System Used in Hovercraft

4.10 Cost Estimation

Cost estimation is important because it helps us know the total amount required to complete the hovercraft project. It also helps in managing the work within budget.

The main cost of this project was related to electronic components, 3D printing material, and assembly items. Motors, ESC units, battery, servo motor, connectors, and RC control parts were the major items used.

We tried to keep the total cost low by selecting suitable and easily available components without affecting the performance of the hovercraft.

The body parts were made using 3D printing, which helped in reducing fabrication cost and provided accurate parts.

Some extra cost was also involved in wiring, screws, and small fitting materials used during assembly.

The detailed cost of components and materials is shown in the table below.

Sr. No.	Components Name	Quantity	Cost
1	BLDC Motor 2212/ 1000KV	1	390.00/-
2	BLDC Motor 2212 /2200KV	1	390.00/-
3	ESC	2	719.98/-
4	Propeller 5045	2	180.02/-
5	Servo MG995	1	260.00/-
6	XT60 Connector	1	40.00/-

From the above cost estimation table, it can be seen that the hovercraft model was completed within a reasonable budget. Most of the cost was related to electronic components and fabrication material. By using 3D printing technology and easily available parts, the total project cost was kept economical.

The final cost of the project was suitable for academic purposes, practical learning, and demonstration work.

4.11 Chapter Summary

In this chapter, we discussed all the main components used in our hovercraft project. Important parts such as **BLDC motors, ESC units, propellers, battery, servo motor, RC control system, connectors,** and **wiring** were explained with their functions in the model.

These components were selected according to the design, power requirement, and weight of the hovercraft. Proper selection of parts helped us achieve lift, forward movement, steering control, and smooth operation.

The estimated project cost was also presented in this chapter. It was found that the hovercraft model was completed within a reasonable budget using easily available components.

Overall, this chapter gave complete information about the materials and components used in making the hovercraft prototype.

CHAPTER 5: WORKING PRINCIPLE

5.1 Introduction

This chapter explains how our RC hovercraft works in practical conditions. The model works by creating a cushion of air below the body, which helps in reducing friction with the ground surface. Due to this, the hovercraft can move smoothly on flat areas.

The working of the model mainly depends on three systems: lift, forward movement, and steering control. All these functions are operated using motors, battery, ESC, servo motor, and remote control components.

When the battery is connected and the system is switched on, power is supplied to both motors through the ESC units. One motor is used for lift generation, while the second motor is used for forward thrust.

The lift motor pushes air below the body and raises the hovercraft slightly above the surface. At the same time, the rear motor pushes air backward, which moves the hovercraft forward.

Direction control is achieved with the help of a rudder connected to a servo motor. When the rudder moves left or right, the hovercraft also changes its direction.

In this way, lift, thrust, and steering work together to make the hovercraft move smoothly and in a controlled manner.

5.2 Lift Generation System

The lift generation system is a very important parts of the hovercraft because it helps the model rise slightly above the surface. Without enough lift, the hovercraft cannot move properly or smoothly.

In our project, lift was generated by using a BLDC motor with a propeller mounted on the body. When the motor starts rotating, the propeller pushes air downward into the lower section of the hovercraft.

This air gets trapped below the body and creates pressure, which forms a cushion of air. As the air pressure increases, the hovercraft rises slightly above the ground surface.

Since the body does not remain in full contact with the ground, friction becomes very low. Due to this, movement becomes easier and the overall performance improves

The lift system was designed carefully so that sufficient air pressure could be created while keeping the body lightweight. Proper fitting and sealing were also necessary to reduce air leakage from the lower section.

During testing, we noticed that lift performance was better on smooth surfaces, where the air cushion could form more effectively.

The lift generation system was very important for keeping the hovercraft stable and allowing smooth movement during operation.

5.3 Thrust Generation System

The thrust generation system is an important part of the hovercraft because it helps the model move in the forward direction. After the hovercraft rises slightly above the surface, thrust is needed to make it travel smoothly.

In our project, we used a BLDC motor at the rear side of the hovercraft. A propeller was connected to the motor shaft. When the motor starts rotating, the propeller pushes air backward with high speed.

As the air moves backward, the hovercraft starts moving forward. This forward force is called thrust. Without enough thrust, the hovercraft may generate lift, but it cannot move properly.

The speed and movement of the hovercraft depend on motor speed, propeller efficiency, battery condition, and total body weight. If the battery is weak or the weight becomes more, the model may move slowly.

The rear motor was fixed carefully inside the housing so that the airflow remains straight during operation. Proper alignment was necessary because uneven airflow can affect balance and movement.

During testing, we observed that the hovercraft moved better when lift and thrust were balanced together. Smooth surfaces also gave better forward motion compared to rough areas.

The thrust generation system was very useful for producing stable forward movement and improving the practical performance of the hovercraft.

5.4 Steering Control System

The steering control system is an important part of the hovercraft because it helps in changing the direction of the model while it is moving. After forward motion begins, steering is required to take turns and guide the hovercraft along the desired path.

In our project, steering was achieved by using a rudder mounted at the rear side of the hovercraft. The rudder was placed in the airflow coming from the rear propeller and was connected to a servo motor operated through the remote control system.

When the user moves the steering control on the transmitter, a signal is sent to the receiver fixed on the hovercraft. The receiver passes this command to the servo motor, which rotates and changes the position of the rudder.

If the rudder moves toward the left side, the airflow is also diverted in that direction and the hovercraft starts turning left. In the same way, movement toward the right side causes the model to turn right.

The turning performance depends on proper rudder size, smooth servo movement, and strong airflow from the thrust motor. If the airflow becomes weak or the rudder gets stuck, the steering response may become slow.

While assembling the system, the rudder position was aligned carefully with the airflow path so that turning remains smooth and controlled.

During practical trials, the hovercraft showed better response when the rudder moved freely and the rear motor was running at suitable speed.

The steering control system was necessary for changing direction, taking smooth turns, and improving handling of the hovercraft.

5.5 Remote Control and Signal Operation

The remote control system is used to operate the hovercraft from a distance. It allows the user to control movement, speed, and turning without touching the model directly.

In our project, we used a transmitter and receiver for this purpose. The transmitter remains in the hand of the user, while the receiver was fixed inside the hovercraft body.

When the user moves the control stick on the transmitter, a signal is send wirelessly to the receiver. The receiver then passes that signal to the required component.

According to the command given by the user, the motors start running faster or slower, and the servo motor changes the rudder position for turning.

Because of this system, the hovercraft can move forward, take left or right turns, and can be controlled easily from a safe distance.

Proper signal range and battery condition are important for smooth working of the remote system. If the signal becomes weak, response may become slow or unstable.

While testing, we found that the hovercraft responded good when the transmitter and receiver were connected properly.

This remote control system made the hovercraft easier to operate and more practical for demonstration purposes.

5.6 Power Supply System

A proper power supply system is necessary for the working of every electrical project, and our hovercraft also depends on it. All major components such as motors, ESC units, receiver, and servo motor require electrical power for their operation.

For this project, we selected a rechargeable LiPo battery as the main source of power. The battery was chosen because it is lightweight, compact in size, and capable of giving sufficient current to run the hovercraft smoothly.

When the battery is connected, power is supplied to the ESC units first. These ESC units then control the speed of both motors according to the signals received from the remote control system.

At the same time, the receiver also gets power for signal communication, and the servo motor receives supply for rudder movement and steering control.

The battery condition has a direct effect on the overall performance of the hovercraft. When the battery is fully charged, lift generation becomes better and forward movement is smoother. If the battery charge becomes low, motor speed reduces and the response of the hovercraft may become weak.

During practical testing, we observed that stable results were obtained when the battery was charged properly before operation.

Proper wiring connections were also important in the system. Loose terminals or poor joints may create power loss and can interrupt the working of motors or control units.

For this reason, the power supply system was one of the key parts responsible for reliable and smooth performance of the hovercraft model.

5.7 Combined Working of Hovercraft

The movement of the hovercraft depends on several systems working together at the same time. A single component cannot run the complete model properly. Lift, thrust, steering, remote control, and power supply all have their own role in the operation of the hovercraft.

When the battery is connected and the remote system is turned on, power is supplied to the receiver, ESC units, motors, and servo motor. After this, the model becomes ready for operation.

The first task is lift generation. The lift motor rotates the propeller and pushes air below the body. This creates an air cushion, which raises the hovercraft slightly above the surface and reduces friction.

Once the model gets enough lift, the rear motor starts producing thrust. The propeller pushes air backward, and as a result the hovercraft begins to move forward.

Direction control is achieved with the help of the rudder. When the user gives a turning command from the transmitter, the servo motor moves the rudder left or right. This changes the airflow direction and the hovercraft turns accordingly.

If all systems work properly together, the model moves in a smooth and stable manner. If battery power becomes low or the body loses balance, movement can be affected.

We noticed that the hovercraft performed better when the weight was distributed properly and both motors were running at suitable speed.

This combined operation helped us understand how mechanical and electronic systems can work together in one practical project.

5.8 Summary

In this chapter, we discussed the complete working principle of the hovercraft and understood how different systems help the model move smoothly. The operation of the hovercraft was mainly based on lift generation, forward thrust, steering control, remote signal operation, and proper power supply.

We studied how the lift motor creates an air cushion below the body and helps the hovercraft rise slightly above the surface. We also understood that the rear motor provides thrust, which is responsible for forward movement of the model.

The steering system was also explained in this chapter. With the help of the rudder and servo motor, the hovercraft can take left and right turns according to the command given by the user through the remote controller.

Another important part was the power supply system, which provides electrical energy to motors, control units, and the steering mechanism. Proper battery condition and wiring were found necessary for better performance.

From this chapter, we learned that the hovercraft works properly only when all systems operate together in balance. If anyone system becomes weak, the overall movement can be affected.

This chapter gave us a clear understanding of the practical working of the hovercraft and created a base for the testing and result analysis discussed in the next chapter.

CHAPTER 6 : FABRICATION, TESTING AND RESULTS

6.1 Introduction

After completing the design and development of the hovercraft, the next step was fabrication, assembly, and practical testing of the model. This stage was important to check whether all designed parts were working properly after manufacturing.

The main aim of this chapter is to explain how the parts were fabricated, how the hovercraft was assembled, and how different tests were carried out to evaluate its performance.

The practical work included 3D printing of designed parts, installation of motors, wiring connections, battery setup, and fitting of steering components.

After assembly, several tests were performed to observe lift generation, forward movement, turning response, balance, and overall stability of the hovercraft.

Some minor problems were also noticed during early trials, which were corrected by making small adjustments in the system.

The observations and results obtained from these practical tests helped us understand the actual working behaviour of the hovercraft.

This chapter presents the complete fabrication process, testing methods, results, and improvements made in the final model.

6.2 Fabrication of Parts

After finishing the CAD design of all required parts, we started the fabrication process of the hovercraft. In this project, most of the components were made by using 3D printing technology because it is convenient for making lightweight and accurate parts in less time.

The main body, skirt frame, rear motor housing, and rudder were first prepared in digital form and then converted into printable files. Each part was checked properly before printing so that fitting problems could be avoided later.

We selected PLA material for printing because it is easy to use, economical, and suitable for prototype models. It also gives a clean finish and enough strength for small project components.

Before starting the print, settings such as layer height, infill, print speed, and support requirement were adjusted according to the shape and size of each part. This was important for getting better print quality.

All parts were printed one by one and removed carefully after completion. Some small finishing work like support removal, edge cleaning, and surface checking was also done where needed.

After printing, each component was checked for size and proper fitting with other parts. Minor corrections were made at some places to make assembly easier.

This fabrication stage was important because it converted our digital design into real physical parts used in the final hovercraft model.

6.3 Final Assembly

After all parts were printed and prepared, we started the final assembly of the hovercraft. In this step, the printed parts and electronic items were joined together to make a complete working model.

First, the main body and skirt frame were fixed properly. These two parts formed the base of the hovercraft, so proper alignment was necessary for good balance and shape.

Next, the lift motor and rear motor were installed in their positions. Care was taken while fitting the motors so that they remain straight and vibration can be reduced.

Then the ESC units, battery, receiver, and wiring were placed inside the body. The components were arranged in a proper way so that weight stays balanced and the inside area remains organized.

After that, the servo motor and rudder were fitted for steering control. Rudder movement was checked by hand before connecting the remote system.

When all parts were fitted, screws, joints, and wire connections were checked once again. Small corrections were made wherever required to avoid problems later.

After completing this work, the hovercraft became ready for practical testing and operation.

This assembly stage was important because all separate parts were converted into one complete working system.

6.4 Testing Procedure

After completing the final assembly, we carried out practical testing of the hovercraft to check its actual working condition. This step was important because it helped us understand whether the model was operating properly and whether all fitted parts were functioning as expected.

Before starting the tests, we charged the battery fully and checked all wire connections carefully. We also inspected motor fitting, rudder movement, receiver response, and the position of mounted components so that unnecessary problems could be avoided.

The hovercraft was tested mainly on a smooth floor surface. We selected a smooth surface because it helps the air cushion form more easily and allows better movement with less friction.

At first, only the lift motor was operated. We observed whether the body was rising slightly above the surface and whether enough air pressure was being created below the hovercraft.

After confirming lift, the rear motor was started to check forward movement. We noted the speed response, smoothness of motion, and overall balance of the model while moving.

Next, steering was tested by giving left and right commands from the transmitter. Rudder movement and turning response were checked carefully to see how well the hovercraft was changing direction.

Several trials were repeated at different speed levels to study stability, control, battery performance, and movement behaviour. Small adjustments were made wherever needed for better results.

All important observations were noted and later used for performance analysis of the developed hovercraft model.

6.5 Performance Results

After completing the testing process, we studied the performance of the hovercraft by observing lift generation, forward movement, turning response, balance, and general working condition of the model. From the trials, we found that the hovercraft was able to perform its main functions properly.

When the lift motor was started, the body of the hovercraft rose slightly above the smooth surface. This showed that the air cushion was being formed below the body and the lift system was working in a proper way.

After that, the rear motor was operated for forward motion. The hovercraft moved ahead smoothly and responded well at normal speed. Better movement was noticed when the battery was fully charged and the weight of the components remained balanced.

The steering system also gave good response. When left or right commands were given from the transmitter, the rudder moved correctly and the hovercraft changed direction without major difficulty.

We also checked the stability of the model while moving. The hovercraft remained reasonably balanced during normal operation. At higher speed, small vibration was noticed, but it was not serious.

The model performed better on smooth and clean surfaces. On rough areas, movement became less smooth because the air cushion was disturbed.

From these practical results, we understood that the developed hovercraft model

6.6 Problems and Improvements

When we started testing the hovercraft, some small problems were noticed in the beginning. This was expected because the model was being tested practically for the first time. After checking the system properly, these issues were improved step by step.

One of the first problems was imbalance of the body. The battery and some internal parts were creating uneven weight, so the hovercraft was not lifting properly at first. We changed the position of the components, and after that the balance became better.

We also noticed slight vibration near the motor section. This happened because the motor fitting was not fully tight in the first setup. After tightening the mounting points, the vibration was reduced.

Another issue was slow turning response at low motor speed. When the rear motor was running slowly, airflow to the rudder became weak, so steering was less effective. Better turning was obtained after increasing the motor speed to a suitable level.

After continuous use, the battery charge also started dropping, which affected lift and forward movement. This was solved by charging the battery again before the next trial.

In some tests, small air leakage from the lower section was also seen. We checked the fitting area and made small adjustments, which helped in improving lift performance.

By making these corrections, the hovercraft started working more smoothly, remained more stable, and gave better control during operation.

6.7 Result Analysis

After completing all the tests and making small improvements, we studied the final performance of the hovercraft carefully. From the practical trials, we found that the model was able to perform the main functions of a small RC hovercraft in a satisfactory way.

The lift system worked properly and created enough air below the body to raise the hovercraft slightly above the surface. Because of this, friction was reduced and movement became easier.

The rear motor also gave good forward movement. The hovercraft moved smoothly at normal speed and responded well to throttle control. At higher speed, slight vibration was noticed, but the model was still controllable.

The steering system gave acceptable results. When commands were given from the transmitter, the rudder moved correctly and the hovercraft was able to take left and right turns without major difficulty.

We observed that the hovercraft performed best on smooth and clean surfaces. On rough areas, movement became less smooth because the air cushion was disturbed.

Battery condition and body balance were also found very important. When the battery was fully charged and the weight was distributed properly, lift and movement became better.

From the final results, we understood that the developed hovercraft was successful as a learning model and clearly demonstrated lift, thrust, steering, and remote control operation in practical use.

6.8 Summary

In this chapter, we discussed the practical work carried out for the hovercraft model, including fabrication, assembly, testing, and final performance analysis. After completing all parts and fitting the required components, the model was made ready for operation.

Different tests were performed to check lift generation, forward movement, steering response, balance, and overall working of the hovercraft. These trials helped us understand the actual behaviour of the model in practical conditions.

Some small problems such as body imbalance, slight vibration, weak turning at low speed, and battery discharge were noticed during initial testing. These issues were improved by making minor adjustments in fitting, weight distribution, and battery charging.

From the final results, we found that the hovercraft was able to rise slightly above the surface, move forward smoothly, and respond properly to steering commands.

This chapter showed that the developed hovercraft was successful as an educational prototype and useful for understanding the practical working of a hovercraft system.

CHAPTER 7: CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

This project was carried out to design and develop a small RC hovercraft using simple fabrication methods and electronic control components. The main purpose of the project was to understand how a hovercraft works in practical conditions and to build a working prototype model.

During the project, we completed different stages such as CAD designing, 3D printing of parts, assembly of components, wiring connections, testing, and small improvements after trials.

The developed hovercraft was able to generate lift, move forward, and respond to steering commands in a proper way. From practical testing, we found that the model performed better on smooth surfaces with proper balance and a fully charged battery.

This project gave us useful knowledge about aerodynamics, mechanical design, electronics, remote control systems, and fabrication methods used in prototype development.

It also helped us improve practical skills such as planning, assembly work, troubleshooting, testing, and solving small problems during the project work.

By completing this project, we understood how mechanical and electronic parts can work together in one system.

The project was successful as a learning model and clearly demonstrated the basic working functions of a hovercraft in a simple and practical way.

7.2 Future Scope

The hovercraft model developed in this project worked well as a basic prototype, but there are still many chances for further improvement in future. By making some changes, the performance and practical use of the model can be increased.

A stronger motor can be used for better lift and smoother forward movement. Battery backup can also be increased so that the hovercraft can run for a longer time without charging again and again.

Control of the model can be made easier by adding Bluetooth, Wi-Fi, or mobile app based operation. This will make the system more modern and convenient to use.

Sensors such as obstacle sensor, gyroscope, and distance sensor can also be added. These sensors may help in better balance, safer movement, and partial automatic control.

A small camera can be fitted on the body for live viewing, inspection work, or surveillance purpose.

The body design can also be improved by using lighter and stronger materials. This may reduce total weight and increase durability of the hovercraft.

In future, with more development, this model can be used in rescue work, research projects, inspection areas, and educational applications.

7.3 Summary

In this chapter, we discussed the final conclusion and future scope of the hovercraft project. From the complete project work, we found that the developed model was successful in showing the basic working principle of a hovercraft in practical form.

The hovercraft was able to generate lift, move forward, and take left and right turns according to control commands. This showed that the lift system, thrust system, and steering system were working properly together.

By completing this project, we gained practical knowledge about CAD design, fabrication, assembly, wiring, remote control operation, testing, and solving small problems during project work.

We also understood that proper body balance, battery condition, and correct fitting of components are important for smooth movement and better performance.

The future scope of the project was also studied. Better motors, improved battery backup, smart control methods, sensors, and camera system can be added later for further development.

This chapter concluded that the hovercraft project was useful as a learning model and can be improved further for advanced practical applications in future.

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4. PTC Creo Parametric User Guide, referred for CAD modelling and design of hovercraft parts.
5. Ultimaker Cura User Manual, referred for slicing settings and 3D printing preparation.
6. Product datasheets of BLDC motor, ESC, servo motor, LiPo battery, propeller, and RC transmitter used in the project.
7. Online technical resources referred for hovercraft working principle, RC control systems, and prototype fabrication methods.
8. Practical observations and test results obtained during fabrication and operation of the developed model.

APPENDIX

LIST OF COMPONENTS, EQUIPMENT AND SOFTWARE USED

1. BLDC Motor (Lift Motor) Quantity: 1 Type: Brushless DC Motor Rating: 1000–1400 KV (Approx.) Use: Air cushion / lift generation	2. BLDC Motor (Thrust Motor) Quantity: 1 Type: Brushless DC Motor Rating: 1000–1400 KV (Approx.) Use: Forward thrust / movement	3. Electronic Speed Controller (ESC) Quantity: 2 Rating: 30A (Approx.) Input: 2S–3S LiPo Compatible Use: Speed control of BLDC motors
4. Servo Motor Quantity: 1 Type: Standard RC Servo Operating Voltage: 4.8V – 6V Use: Rudder movement for steering	5. RC Transmitter Quantity: 1 Frequency: 2.4 GHz Channels: 2 or more Use: Wireless control of hovercraft	6. RC Receiver Quantity: 1 Frequency: 2.4 GHz Channels: Matching transmitter Use: Receives control signals from
7. LiPo Battery Quantity: 1 Type: Rechargeable Lithium Polymer Voltage: 11.1V (3S) Capacity: 2200 mAh (Approx.) Use: Main power supply	8. Propellers Quantity: 2 Material: Plastic / Nylon Size: 3 Inch Use: Lift and thrust airflow generation	9. PLA Filament Quantity: As required Diameter: 1.75 mm Use: 3D printing of body parts

SOFTWARE USED

11. FreeCAD Version: Latest Available Version Use: CAD modelling of hovercraft parts	12. Ultimaker Cura Version: 5.10.2 Use: Slicing of 3D models and Stl.file generation
13. Microsoft Word Version: As Installed Use: Report writing and documentation	